
Notes for ‘The Fundamental Theorem of Calculus and indefinite integrals’

Important Ideas and Useful Facts:

- (i) **Antiderivatives and Uniqueness up to a Constant:** If f and g are functions such that f is differentiable and $f' = g$ then we call g the *derivative* of f and f an *antiderivative* of g .

If f_1 and f_2 are antiderivatives of g , and g is continuous on an interval, then the values of f_1 and f_2 differ by a constant on this interval, that is, there exists a constant C such that

$$f_1(x) = f_2(x) + C$$

for all x in the interval. We call C the *constant of integration*.

- (ii) **The Fundamental Theorem of Calculus:** If $a < b$ and f is continuous for $a \leq x \leq b$, and F is any antiderivative of f for $a \leq x \leq b$, then

$$\int_a^b f(x) dx = F(b) - F(a) .$$

Common notations for $F(b) - F(a)$ are $[F(x)]_a^b$, $F(x)|_a^b$ and $F(x)|_a^b$.

- (iii) **The Indefinite Integral:** If f is a continuous function then we write

$$\int f(x) dx$$

for any antiderivative of f , and call this the *indefinite integral* of f . Thus if we put $F(x) = \int f(x) dx$, for some choice of antiderivative, then $F'(x) = f(x)$. Choices of antiderivatives differ by a constant.

- (iv) **Some Properties and Standard Indefinite Integrals:**

$$(a) \quad \int k f(x) dx = k \int f(x) dx \quad \text{and} \quad \int f(x) + g(x) dx = \int f(x) dx + \int g(x) dx .$$

$$(b) \quad \int x^n dx = \frac{x^{n+1}}{n+1} + C \quad \text{for } n \neq -1, \text{ and } \int \frac{1}{x} dx = \ln|x| + C .$$

$$(c) \quad \int e^x dx = e^x + C, \quad \int \ln x dx = x \ln x - x + C .$$

$$(d) \quad \int \cos x dx = \sin x + C, \quad \int \sin x dx = -\cos x + C, \quad \int \sec^2 x dx = \tan x + C .$$

Examples and derivations:

1. Find the indefinite integral $\int (6x^2 + 4x - 1) dx$.

Solution: We can integrate each piece separately, and combine the constants as a single “plus C ” at the right, to get

$$\begin{aligned}\int (6x^2 + 4x - 1) dx &= 6 \int x^2 dx + 4 \int x dx - \int 1 dx \\ &= 6 \left(\frac{x^3}{3} \right) + 4 \left(\frac{x^2}{2} \right) - x + C \\ &= 2x^3 + 2x^2 - x + C.\end{aligned}$$

In practice, one usually goes immediately to the last step, adjusting the constants quickly as one goes.

2. Find the definite integral $\int_1^2 (6x^2 + 4x - 1) dx$.

Solution: Using the antiderivative from the previous exercise (without the “plus C ”), and the Fundamental Theorem of Calculus, we get

$$\int_1^2 (6x^2 + 4x - 1) dx = [2x^3 + 2x^2 - x]_1^2 = 16 + 8 - 2 - (2 + 2 - 1) = 19.$$

3. Find the indefinite integral $\int (\cos x - \sin x) dx$.

Solution: Again we can integrate each piece separately, and combine the constants as a single “plus C ” at the right, to get

$$\int (\cos x - \sin x) dx = \int \cos x dx - \int \sin x dx = \sin x + \cos x + C.$$

4. Find the definite integral $\int_{\pi/4}^{\pi/2} (\cos x - \sin x) dx$.

Solution: Using the result of the previous exercise, we have

$$\begin{aligned}\int_{\pi/4}^{\pi/2} (\cos x - \sin x) dx &= [\sin x + \cos x]_{\pi/4}^{\pi/2} = \sin \frac{\pi}{2} + \cos \frac{\pi}{2} - \left(\sin \frac{\pi}{4} + \cos \frac{\pi}{4} \right) \\ &= 1 + 0 - \left(\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} \right) = 1 - \sqrt{2}.\end{aligned}$$

5. Find the indefinite integral $\int 2x\sqrt{x} dx$.

Solution: We rewrite the integrand as a single fractional power of x , and as usual rewrite any constant to the right as a “plus C ”, to get

$$\int 2x\sqrt{x} dx = 2 \int x^{3/2} dx = 2 \left(\frac{x^{5/2}}{5/2} \right) + C = \frac{4x^{5/2}}{5} + C.$$

6. Find the definite integral $\int_1^4 2x\sqrt{x} \, dx$.

Solution: Using the result of the previous exercise, we have

$$\int_1^4 2x\sqrt{x} \, dx = \left[\frac{4x^{5/2}}{5} \right]_1^4 = \frac{4}{5} [x^{5/2}]_1^4 = \frac{4}{5} (32 - 1) = \frac{124}{5} .$$

7. Find the definite integral $\int_1^2 (x^{-2} - 2x^{-3}) \, dx$.

Solution: By the formula for integrating powers of x for each of the two pieces, we get

$$\int_1^2 (x^{-2} - 2x^{-3}) \, dx = [-x^{-1} + x^{-2}]_1^2 = \left(-\frac{1}{2} + \frac{1}{4} \right) - (-1 + 1) = -\frac{1}{4} .$$

8. Find the definite integral $\int_{-1}^1 \frac{e^x - e^{-x}}{2} \, dx$.

Solution: An obvious antiderivative of e^x is e^x . For e^{-x} , it is not so hard to guess an antiderivative, because its derivative is $-e^{-x}$, so multiplying through by negative one, gives e^{-x} . Hence, considering the indefinite integral first, and playing with minus signs for e^{-x} , we get

$$\int \frac{e^x - e^{-x}}{2} \, dx = \frac{1}{2} \left(\int e^x \, dx - \int e^{-x} \, dx \right) = \frac{1}{2} (e^x + e^{-x}) + C .$$

Hence, we get

$$\int_{-1}^1 \frac{e^x - e^{-x}}{2} \, dx = \frac{1}{2} [e^x + e^{-x}]_{-1}^1 = \frac{1}{2} (e + e^{-1} - e^{-1} - e) = 0 .$$

9. Suppose that F and G are antiderivatives of f on some interval. We sketch a proof why there exists a constant C such that

$$F(x) = G(x) + C$$

for all x on that interval.

Sketch of proof: We have that

$$F'(x) = G'(x) = f(x)$$

for all x in the interval. The trick is to look at the function H defined by the rule

$$H(x) = F(x) - G(x) .$$

Then

$$H'(x) = F'(x) - G'(x) = f(x) - f(x) = 0 ,$$

so that the tangent lines to the graph of H are all horizontal. Intuitively, it is clear that this forces the graph of H itself to be a horizontal line, for if the curve started to move away from the horizontal at some point, then it could not be approximated arbitrarily well by horizontal tangent lines. Hence $H(x)$ is constant for all x in the interval, so

$$F(x) - G(x) = H(x) = C ,$$

for some constant C , so that

$$F(x) = G(x) + C ,$$

completing the sketch of the proof.

10. We explain why there is a magnitude sign appearing in the standard integration formula

$$\int \frac{dx}{x} = \ln|x| + C .$$

For $x > 0$, certainly $\ln x$ is an antiderivative of $\frac{1}{x}$, so there is some constant C_1 such that

$$\int \frac{dx}{x} = \ln x + C_1 . \quad (1)$$

However, let us now consider $x < 0$. Then $-x > 0$ and the expression $\ln(-x)$ makes perfectly good sense. Put $u = -x$, so that $\frac{du}{dx} = -1$. By the Chain Rule,

$$\frac{d}{dx}(\ln(-x)) = \frac{d}{du}(\ln u) \frac{du}{dx} = \left(\frac{1}{u}\right)(-1) = \left(\frac{1}{-x}\right)(-1) = \frac{1}{x} .$$

So, in this case, $\ln(-x)$ is an antiderivative of $\frac{1}{x}$, so there must be some constant C_2 such that

$$\int \frac{dx}{x} = \ln(-x) + C_2 . \quad (2)$$

If $C_1 = C_2 = C$ then (1) and (2) can be combined in a single equation,

$$\int \frac{dx}{x} = \ln|x| + C , \quad (3)$$

exploiting the fact that $|x| = x$ for $x > 0$, and $|x| = -x$ for $x < 0$. This explains the usual formula.

In fact, (1) and (2) are more general than (3), because one could create a single antiderivative of $y = \frac{1}{x}$ using different constants in (1) and (2) for the different branches of the hyperbola, for example

$$F(x) = \begin{cases} \ln x + 3 & \text{if } x > 0 \\ \ln(-x) + 7 & \text{if } x < 0 \end{cases}$$

which would not be covered by (3). However, in practice, one is usually only concerned with one branch of the hyperbola, where the domain is a connected interval, so (3) suffices. Note that, because of this flexibility on the different branches of the hyperbola, the result guaranteeing uniqueness of the antiderivative up to addition by a single constant fails. This is not a contradiction, because the function $y = \frac{1}{x}$ is not continuous on the whole real line.